AP PRECALCULUS

Unit 1 • Lesson 1.10 practice

Lesson 1.10 • Rational Functions and Holes

4 multiple-choice questions and 1 free-response question

Question 1

Where is the hole in the graph of $r(x) = \dfrac{x^2 - 25}{x - 5}$?

(A) $(5, 0)$
(B) $(5, 10)$
(C) $(-5, 0)$
(D) $(5, 5)$

Question 2

Where is the hole in the graph of $r(x) = \dfrac{x^2 + x - 2}{x^2 - 1}$?

(A) $\left(1, \frac{3}{2}\right)$
(B) $(-1, 0)$
(C) $(1, 0)$
(D) $(-2, 0)$

Question 3

What happens on the graph of $r(x) = \dfrac{x - 2}{(x - 2)^2}$ at $x = 2$?

(A) A hole
(B) A zero
(C) A vertical asymptote
(D) Nothing; r is defined at $x = 2$.

Question 4

$\lim\limits_{x \to 3} \dfrac{x^2 - 3x}{x - 3}$ is

(A) 0
(B) 1
(C) 9
(D) 3

Question 5 (Free response)

Let $r(x) = \dfrac{x^3 - 4x}{x^2 + 2x}$.

(a) Factor $r(x)$ completely and simplify.
(b) Find the coordinates of every hole in the graph of r.
(c) Does the graph of r have a vertical asymptote? Find the zeros of r and describe the graph.